

Constructing J

$$J(X) = \frac{\partial f}{\partial X}(X)$$

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Constructing A

$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

$$A = \tilde{U}^T J \tilde{V}$$

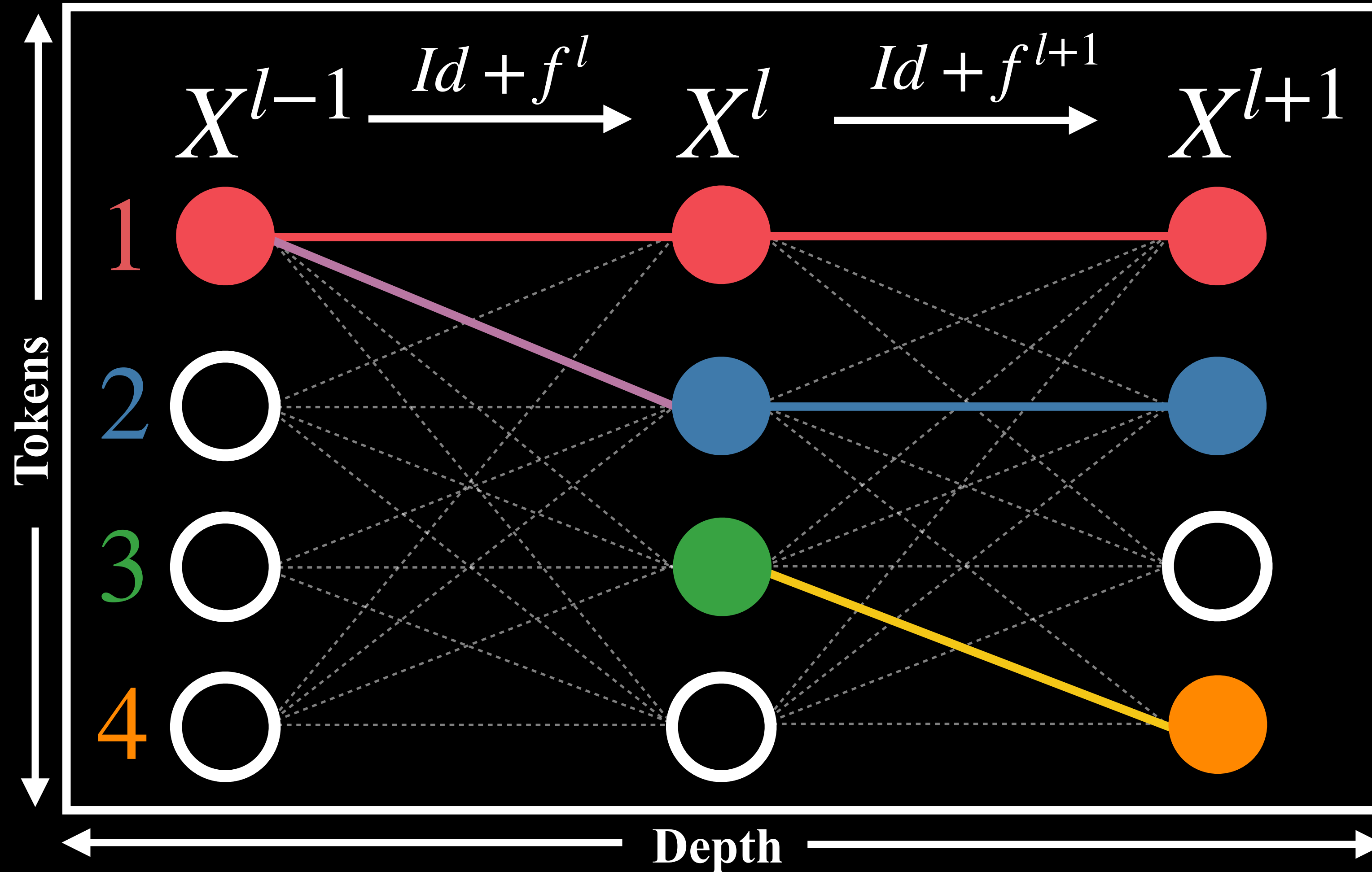
$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

$$A = \tilde{U}^T J \tilde{V}$$

Depth-wise Coupling

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} \left(f^l(X^{l-1}) \right)_1 \\ J_{11}^{l+1} &= \frac{\partial}{\partial x_1^l} \left(f^{l+1}(X^l) \right)_1 \end{aligned} \right\} A_{11}^{l,l+1}$$

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1 \\ J_{11}^{l+1} &= \frac{\partial}{\partial x_1^l} (f^{l+1}(X^l))_1 \end{aligned} \right\} A_1^{l,l+1}$$



Token-wise Coupling

Input = Output

$$J_{11}^l = \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1$$

$$J_{22}^{l+1} = \frac{\partial}{\partial x_2^l} (f^{l+1}(X^l))_2$$

}

$A_{12}^{l,l+1}$

Input $\neq$ Output

$$J_{12}^l = \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_2$$

$$J_{34}^{l+1} = \frac{\partial}{\partial x_3^l} (f^{l+1}(X^l))_4$$

}

$A_{1234}^{l,l+1}$

Input = Output

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1 \\ J_{22}^{l+1} &= \frac{\partial}{\partial x_2^l} (f^{l+1}(X^l))_2 \end{aligned} \right\} A_{12}^{l,l+1}$$

Input $\neq$ Output

$$\left. \begin{aligned} J_{12}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_2 \\ J_{34}^{l+1} &= \frac{\partial}{\partial x_3^l} (f^{l+1}(X^l))_4 \end{aligned} \right\} A_{1234}^{l,l+1}$$

Visualization of A

Randomly Initialized

GPT

